

Operator Manual

This manual covers running AirOne during a mission or a demonstration.

1. Starting the ground station

All commands are run from the `airone/` directory.

1.1 Prerequisites

`pip install -r requirements.txt`

`export AIRONE_JWT_SECRET="<a long random secret, >= 32 chars>"`

The `AIRONE_JWT_SECRET` environment variable is **mandatory**. The system refuses to start with a default/placeholder secret — this is a deliberate security control, not a bug (see [security_model.md](#)).

1.2 Launch modes

Command	What it does
<code>python launcher.py</code>	Starts the backend (API + workers). Reads live telemetry from the configured serial port.
<code>python launcher.py --simulate</code>	Feeds a physics-based simulated mission through the real pipeline. All resulting data is tagged SIMULATED.
<code>python launcher.py --simulate --sim-loop</code>	Continuously replays the simulated mission (useful for demos).
<code>python launcher.py --gui</code>	Also starts the PyQt5 operator GUI (if a display and PyQt5 are available).
<code>python launcher.py --validate-only</code>	Validates configuration and environment, then exits without starting.

Simulation tuning flags: `--sim-seed <int>`, `--sim-apogee <metres>`, `--sim-rate <Hz>`.

Honesty note: Data produced in `--simulate` mode is never presented as real. Every measurement carries `source = SIMULATED` and the GUI shows a simulation banner. This is how you demo the system without ever faking a real reading.

1.3 Accounts

No accounts ship with the software. Create them with the launcher (the password comes from `AIRONE_ADMIN_PASSWORD` or is generated and printed once):

```
AIRONE_ADMIN_PASSWORD='<strong password>' python launcher.py --create-admin  
python launcher.py --create-user ops1 --role OPERATOR --generate-password  
python launcher.py --reset-password ops1 --must-change  
python launcher.py --list-users
```

Passwords need 12+ characters with upper/lower/digit/symbol. Five wrong attempts lock the account for 5 minutes (doubling on repeat); the API answers 429 ACCOUNT_LOCKED with Retry-After. A user flagged *must change* gets 403 PASSWORD_CHANGE_REQUIRED until POST /api/v1/auth/change-password.

The GUI logs in with AIRONE_GUI_USER / AIRONE_GUI_PASSWORD; without them it runs UNAUTHENTICATED and shows only public status.

python launcher.py --verify-audit checks the audit log hash chain (exit 0 = intact, 2 = broken, reports the first bad line).

2. The GUI

The GUI is a **decoupled REST client** of the backend API. It runs in the main thread; the backend runs in worker threads. A GUI crash therefore cannot corrupt telemetry capture — the two are isolated by design. In a headless environment (--gui with no DISPLAY), the launcher reports this honestly and continues running the backend without a window.

2.1 Tabs

1. **Mission Control** — live mission state, latest telemetry, mission-state controls.
2. **Science Dashboard** — the 37 scientific analyses with values, units, and uncertainty.
3. **AI Diagnostics** — advisory ML anomaly detection, drift, and the model registry.
4. **Map** — GNSS ground track (requires valid GNSS fixes).
5. **Replay** — step through recorded telemetry history.
6. **Presentation / Judge Mode** — a clean, large-format read-only view for demos.
7. **Configuration** — view/edit runtime configuration (ENGINEER role required).

2.2 Colour vocabulary

The GUI colours every value by its quality state. A red/orange cell is not a styling choice — it means the data is INVALID, STALE, or SUSPECT and must not be trusted. Never read a number without reading its colour.

3. Judge / Presentation mode

Presentation mode is designed for scored demonstrations:

- Large fonts, high-contrast theme, no editable controls.
- Shows only data the system can currently stand behind. If a value is stale or a tier is down, the panel says so rather than showing an old number.
- A persistent banner shows whether the session is LIVE or SIMULATED.

4. Exporting data

Telemetry can be exported via the API (GET /api/v1/telemetry/export, OPERATOR role, rate-limited to 5/minute). Exports include the quality state of every measurement so downstream consumers inherit the same honesty guarantees.

5. Shutting down

Ctrl-C triggers a graceful shutdown: workers are stopped in reverse dependency order, in-flight telemetry is flushed to the database, and a final shutdown-state snapshot is written to data/last_shutdown_state.json.